



# Draft genome analysis of a multidrug-resistant *Pseudomonas aeruginosa* CMPL223 from hospital wastewater in Dhaka, Bangladesh

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## ABSTRACT

**Objectives:** Multidrug-resistant (MDR) clones of *Pseudomonas aeruginosa* can cause complicated infections in human. The emergence of ST664 of MDR *P. aeruginosa* has been reported in Nepal, Iran and China. Here, we present the draft genome analysis of a MDR *P. aeruginosa* CMPL223 isolated from hospital wastewater in Bangladesh to understand antimicrobial resistance trends and pathogenicity.

**Methods:** Cetrimide agar was used for isolation of *P. aeruginosa*. Polymerase chain reaction (PCR) was carried out for detection of biofilm and integron related genes. Bacterial susceptibility to antibiotics was determined by disc diffusion method. Sequencing of whole genomic DNA was performed using Illumina iSeq 100 platform. Following quality checking of raw reads, assembly and annotation of sequences, a wide array of in silico tools were used for characterization of draft genome.

**Results:** The isolate was a strong biofilm former, carried integron 1 in chromosomal DNA, and was predicted to be pathogenic. It belongs to sequence type ST664 and O7 serogroup. The assembled genome contained 12 acquired antimicrobial resistant (AMR) genes, 2 prophage regions, 240 virulence genes, 71 drug targets, 142 insertion sequences, and 1 CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) array. The isolate was resistant to 21 out of 23 antibiotics, except colistin and imipenem. Comprehensive Antibiotic Resistance Database and ResFinder revealed that bacteria harboured *bla*<sub>OXA-50</sub>, *bla*<sub>OXA-796</sub>, *bla*<sub>PPC-374</sub>, *fosA*, *tet*(G), *sul*1, *cat*B7, *aph*(3')-iib and *ant*(4')-IIb genes, conferring resistance to different classes of antibiotics. The results of in vitro characterization were consistent with the possible expression of detected antibiotic resistant genes through in silico analysis.

**Conclusion:** Our data suggested the emergence of MDR *P. aeruginosa* ST664, which needs control measures for limiting its dissemination.

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## 1. Introduction

Antibiotics are frequently used in hospitals to cure bacterial infections. Continuous discharge of antibiotics in patient excrement makes hospital effluent a reservoir of a large amount of residual antibiotics and bacteria that exerts selective pressure on the emergence of antibiotic-resistant bacteria [1]. Many studies have reported that multidrug-resistant (MDR) *Pseudomonas aeruginosa* can persist in wastewater over extended periods and acquire numerous resistance genes to different classes of antibiotics through horizontal gene transfer [2,3]. Carbapenem-resistant *P. aeruginosa* is in-

creasingly prevalent in hospital environments in Bangladesh with no effective antibiotics for infection management; it thus poses a serious threat to public health [4]. Proper understanding of the evolution and transmission of resistant bacteria is obvious for making effective interventions. Therefore, *P. aeruginosa* CMPL223 was analysed by whole-genome sequencing (WGS) to keep track of the spread of newly emerging virulent and MDR strains.

## 2. Material and methods

A 100-mL sample of hospital wastewater was collected from the central wastewater discharge point of a hospital in Dhaka, Bangladesh in September 2021, and 100 µL of dilutions (10<sup>-1</sup> to 10<sup>-5</sup>) of samples were spread on Cetrimide agar to isolate *P. aeruginosa* presumptively identified by a panel of biochemical tests.

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Bacterial biofilm forming ability was measured by Crystal Violet microtiter plate assay. The steps included adjustment of the turbidity of bacterial culture to 1 MacFarland standard, inoculation of 200 µl culture on a flat-bottom 96-well microtiter plate, incubation at 37 °C for 24 h, washing out planktonic cells with sterile phosphate buffered saline (PBS), staining biofilm with 200 µl 0.1% Crystal Violet for 5 min, washing again to remove stain, solubilization of remaining stain in 200 µl 96% ethanol and, finally, taking absorbance at 492 nm using a microplate reader. The OD<sub>492</sub> value and multiples of the cut-off OD value (OD<sub>c</sub> = 0.06) were used to measure biofilm development (strong biofilm formation when OD > 4 × OD<sub>c</sub>) [5]. Moreover, gene-specific PCR was carried out for molecular detection of biofilm-forming genes (*algD*, *pslD*, *pelf*) and integron (*int1*, *int2*, *int3*) in chromosomal DNA. The Kirby Bauer disc diffusion method was followed for determining antibiotic susceptibility on Mueller Hinton agar (MHA) using the following discs: amikacin (30 µg), streptomycin (10 µg), gentamicin (10 µg), tobramycin (10 µg), netilmicin (10 µg), ceftazidime (30 µg), cefepime (30 µg), cephadrine (30 µg), cefazolin (30 µg), aztreonam (30 µg), colistin (10 µg), tetracycline (30 µg), imipenem (10 µg), meropenem (10 µg), sulphamethoxazole (25 µg), ciprofloxacin (5 µg), ofloxacin (5 µg), levofloxacin (5 µg), amoxicillin + clavulanic acid (30 µg), ampicillin (10 µg), erythromycin (10 µg), teicoplanin (30 µg) and chloramphenicol (30 µg). Briefly, the isolate was grown on Mueller Hinton broth and turbidity was adjusted to 0.5 MacFarland standard. Uniform lawn was prepared by spreading culture on MHA using a sterile cotton swab, and antibiotic discs were placed on the surface of the media. Following incubation at 37 °C for 24 h, the diameter of the zone of inhibition was measured and Clinical Laboratory Standards Institute (CLSI) guideline was followed to classify the isolate as sensitive, intermediate or resistant.

Genomic DNA was extracted using a Wizard Genomic DNA Purification Kit. A Nextera DNA Flex Library Preparation kit was used for sequencing library preparation followed by sequencing on the Illumin iSeq 100 platform. FastQC (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc>), Trimmomatic [6] and SPAdes [7] were employed for quality assessment of raw data, adapter trimming and sequence assembly, respectively. The assembled genome was uploaded at the Type Strain Genome Server [8] for taxonomic identification and phylogenetic inference.

Genome annotation was done using rapid annotation using Subsystem Technology 2.0 (RAST) (<https://rast.nmpdr.org>) and PATRIC 3.6.12 (<https://www.patricbrc.org>). The Comprehensive Antibiotic Resistance Database (<https://card.mcmaster.ca>), ResFinder 4.1 (<https://cge.cbs.dtu.dk/services/ResFinder>) and ABRicate (<https://github.com/tseemann/abricate>) were used for in silico prediction of AMR genes. The assembled genome was submitted to PathogenFinder 1.1 (<https://cge.food.dtu.dk/services/PathogenFinder>), ISfinder (<https://isfinder.biotoul.fr>), MLST 2.0 (<https://cge.food.dtu.dk/services/MLST>), Past 1.0 (<https://cge.food.dtu.dk/services/Past>), CRISPRfinder (<https://crispr.i2bc.paris-saclay.fr/Server>), VFDB (<http://www.mgc.ac.cn/VFs>), PHASTER (<https://phaster.ca>), antiSMASH (<https://antismash.secondarymetabolites.org/#!/start>) and BAGEL4 (<http://bagel4.molgenrug.nl>) to identify pathogenic level, insertion sequences, sequence type (STs), serogroup, CRISPR array, virulence gene, prophage sequences, secondary metabolites and bacteriocin, respectively.

### 3. Results

The size of the assembled genome was 6 861 544 bp, which possessed 473 (≥500 bp) contigs with 66% GC content, N50 = 35689, L50 = 63, 6808 coding sequences (CDS), 46 tRNA and 3 rRNA. Whole genome comparison-based taxonomy identified the CMPL223 as *P. aeruginosa*, and phylogenetic analysis revealed

*P. aeruginosa* DSM 50071 as its closest relative (Supplementary Fig. 1). The isolate was predicted to be pathogenic (probability of being a human pathogen = 0.74 out of 1) with a strong biofilm-forming ability (OD = 0.410). According to PCR results, the isolate carried all the biofilm-forming genes and only class 1 integron. Moreover, the isolate was found to be resistant to 21 out of 23 tested antibiotics (91%) except colistin and imipenem.

Based on the clustering of housekeeping genes (*acsA*, *aroE*, *guaA*, *mutL*, *nuoD*, *ppsA*, *trpE*) with human strains, in silico analysis of the assembled genome indicated that this strain belongs to ST664. It also belongs to O7 serogroup according to PAST-1.0, a highly efficient in silico serotyping tool that predicts serogroup based on the sequence of the O-specific antigen (OSA) gene cluster. The strain carried seven prophage regions including two intact (node 54, 70) and five incompletes (node 78, 172, 189, 196, 200). Two intact regions (score = 150) had resemblance to prophage Pseudo\_PMG1\_NC\_016765 and Pseudo\_PhiCTX\_NC\_003278; 142 insertion sequences, notably (e value ≤0.5) ISPsy43 from IS66 family, IS900 from IS100 family, IS987, IS986, IS986 from IS3 family and ISAau4 from Tn3 family, were distributed into the genome. Analyses of the strain for the CRISPR-Cas system revealed one CRISPR array at contig 359 (CRISPR position 16–180, length 164 bp, DR length = 23, number of spacers = 3). Genome annotation using antiSMASH and BAGEL4 predicted the gene cluster for several antimicrobial compounds such as 2-amino-4-methoxy-trans-3-butenic acid, bottromycin, pyocins2 and scatipeptides. It also carried a number of pathways for degradation and metabolism of xenobiotics such as DDT, caprolactam, atrazine, ethylbenzene, naphthalene, 2,4-dichlorobenzoate, fluorobenzoate, bisphenol A, geraniol and others.

The isolate possessed an arsenal of virulence determinants involved in adherence (*flagella*, *motA*; pili biosynthesis, *fimT*, *pilC*; twitching motility), antimicrobial activity (phenazine biosynthesis, *phzA*, *phzM*, *phzS*), antiphagocytosis (alginate biosynthesis and regulation, *algD*, *algR*, *mucC*), biosurfactant (rhamnolipid biosynthesis, *rhl*), enzyme and protein (hemolytic phospholipase C, *plcH*; phospholipase D, *pldA*; elastase, *lasA*, *lasB*; alkaline protease, *aprA*), secretion system (TTSS translocated effectors, *exoS*, *exoU*; type II secretion system, type III secretion system, type VI secretion system), iron uptake (achromobactin biosynthesis, *acsA*; pyochelin, *pch*; pyoverdine, *pvd*), quorum sensing (*hdtS*, *lasR*) (Supplementary Table 1).

A total of 12 acquired antibiotic resistance genes were detected in CMPL223 through the ResFinder database that conferred resistance to aminoglycosides, fluoroquinolone, cephalosporin, sulfonamide, tetracycline, beta-lactam, chloramphenicol, etc. Detailed information on all of the AMR genes, their respective products and drug class are enlisted in Supplementary Table 4. Some mentionable products of these genes were Class C and D beta-lactamase (*bla*<sub>pdC-374</sub>, *bla*<sub>OXA-50</sub>, *bla*<sub>OXA-796</sub>), aminoglycoside 3'-phosphotransferase (*aph*(3')-iib), chloramphenicol/florfenicol efflux MFS transporter FloR2, aminoglycoside 4'-nucleotidyltransferase (*ant*(4')-IIb), chloramphenicol acetyltransferase (*catB7*), ciprofloxacin resistance protein (*CrpP*), sulfonamide-resistant dihydropteroate synthase (*Sul1*) and others. These genes confer resistance by antibiotic inactivation, target replacement, antibiotic efflux and other mechanisms. A circular genome map showing some important virulence and AMR genes and other genomic features are presented in Fig. 1.

### 4. Discussion

The global emergence and dissemination of novel sequence types and clones of antibiotic-resistant *P. aeruginosa* with a significant epidemic risk in hospitals have drawn immediate attention in a number of recent reports [9]. Therefore, we conducted the



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